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Viewpoint

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Public Administration Challenges in the World of AI and Bots

Abstract: *Technology-driven disruption is taking place at a pace and scale not witnessed before in history. Waves of technology, such as the internet of things, big data, machine learning, and artificial intelligence, are reshaping our personal and professional lives in profound ways. A new world is emerging in which many of the current job classes will disappear, while new ones, requiring entirely different sets of skills, are emerging. Public administrators are unprepared for the challenges they must face in order to cope with this nonincremental and exponential change. Many of the existing government structures and processes that have evolved over the last few centuries will likely become irrelevant in the near future. There is a compelling need to lay the groundwork for governments to rethink how they will be able to best serve their constituents.*

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"Dispense with a horse and save the expense, care and anxiety of keeping it . . ."—thus was the value proposition expressed in one of the first car advertisements. (ca. 1898)

In 1870, Siegfried Marcus built the first gasoline-powered combustion engine, followed by Karl Benz, who developed the first motorized vehicle in 1885. Thirty years later, in 1915, the Ford Motor Company passed the production milestone of one million vehicles, and by 1927, that company alone had produced more than 15 million cars. The shadow of the horse loomed on in this new world, and to this day, it is no accident that we measure vehicle engines in *horsepower*.

The car was changing our roads, but away from our cities, tractors had started to appear on the scene and replace horses on American farms. In 1917, Ford introduced the Fordson tractor, which could be considered the iPad of tractors—easy to use and very productive. Thus began a sharp, long-term decline in the number of horses used on American farms and in cities and a corresponding exponential increase in the numbers of tractors and cars.

Over the next few decades, life became increasingly difficult for those in the horse business, whether on farms or in cities. Agricultural employment decreased significantly, and the unemployed lacked skills relevant for the new economy. World War I temporarily alleviated the job crisis, but it was not until the New Deal that this unemployment issue was seriously addressed through public policy. Such is the nature of disruption that comes with innovation. Public policy is reactive by its very nature. It is very difficult to make a case and find champions for problems that do not yet exist.



Figure 1 Car Advertisement, ca. 1898

For much of human history, horses were our primary source of power and transportation. Environmentalists among you may see the horse as the pinnacle of clean technology, but New York City alone had 100,000 horses, producing 2.5 million pounds of manure a day. The *Times* of London in 1894 forecast that by the 1950s, London would be buried under 9 feet of manure.

Fast-forward to the present, a couple of decades into the twenty-first century: we are witnessing a very similar social and economic transformation that is taking place as a result of waves of disruptive technologies sweeping our landscape. The industrial era saw horses lose their jobs—is this century the one in which we do? Headlines across all platforms are proclaiming the takeover of robots, machine learning, and artificial intelligence (AI). However, the rapid rate of current change is much more severe in comparison with the Industrial Revolution. According to McKinsey Global Institute, “we estimate that this change is happening ten times faster and at 300 times the scale, or roughly 3,000 times the impact” (Dobbs, Manyika, and Woetzel 2015). An impending change of this magnitude poses a formidable challenge for public administrators. There is a compelling need to act sooner rather than later.

The Waves of Technology

Beginning with mainframes in the 1960s, we have witnessed wave after wave of technological innovation. The mainframe wave was followed by mini-computers, then PCs and networking.

The convergence and maturity of these technologies brought us the internet in the 1990s, beginning as the internet of documents, followed by the internet of commerce, the internet of people (social media), and, more recently, the internet of things (IoT). Each wave is built on the momentum created by the one preceding it, and thus each flow of innovations carries greater power to penetrate our lives. All of these innovations are driven by three fundamental technology drivers: continuous exponential reduction in the costs of computing, storage, and connectivity. Furthermore, the infotech revolution is supporting the cleantech and biotech revolutions, creating disruptions in their own spheres.

The first wave of the internet of documents gave us web pages and email, with the number of people connected to the internet in the hundreds of millions. The next wave, the internet of commerce, gave us online shopping, increasing the number of people on the internet by a factor of three to five. The internet of people or social media has now connected more than half of humanity. The IoT wave will connect 20 billion to 30 billion things over the next three to four years. The IoT wave is creating a new world in which every physical object capable of hosting a sensor and using power and connectivity will be connected to the internet. IoT is shaping a new world in which our objects will talk to us.

Consumer IoT is creating gadgets to make our lives more interesting. Thermostats, doorbells, garage doors, and chairs are all being connected. Some unusual examples of IoT connectivity include a connected toothbrush that keeps track of your dental hygiene habits, an internet-connected refrigerator that allows you to look inside your refrigerator while you are at the supermarket, and even IoT pet toys enabling you to monitor your cat's exercise habits.

The use of IoT in the public sphere is dubbed “smart city.” A smart city utilizes and deploys IoT to manage transportation, power plants, water supply, waste management, law enforcement, and so on, in the process optimizing a city's operations.

The use of IoT in the industrial sector is labeled industrial IoT or IIoT. IIoT is the use of IoT technologies in creating connectivity in the machinery and vehicles required in the transport, energy, and other industrial sectors. The opportunities to create efficiencies from IIoT are enormous. Imagine an IIoT-enabled jet engine telling you when it is time for maintenance or repair instead of following predesigned schedules, thus creating considerable operating efficiencies and increasing passenger safety.

These 20 billion to 30 billion connected devices will generate massive amounts of data far beyond the comprehension of any group of human beings. Machine learning and AI will be critical in making sense of all this data and improving decision making. Clive Humby, an industry pioneer, coined the phrase “data is the new oil,” an apt metaphor for how data will drive machine learning. Data, however—unlike oil, which is a limited natural resource—is limitless and bound only by our ability to store and process it.

The Growing Income Disparity

Disruptions upset the status quo and create new opportunities that entrepreneurs capitalize on, thereby creating new wealth. To some extent, this helps explain the income gap and the shrinking middle-class phenomenon that we currently face. We saw the emergence of “barons” during the Industrial Revolution. Similarly, the current tech-driven disruptions are creating massive amounts of new wealth, but it is concentrated among a few entrepreneurs who are willing to take risks and those with skills that support the disruptive industries and ventures.

To give an idea of the magnitude of this new wealth creation, the market cap of tech or tech-driven companies that have gone public in the last 35 years is in the range of \$15 trillion to \$20 trillion. By comparison, all of the gold in the world is valued at \$8 trillion; the size of the U.S. economy is approximately \$20 trillion. The market cap of just six companies—Amazon, Apple, Facebook, Google, Microsoft, and Netflix—is in excess of \$3.5 trillion. At the individual level, of the top 10 richest men in the United States (sometimes referred to as the three-comma club, since a billion has three commas), seven have made their wealth in the tech or information sector. The collective net worth of those seven is in excess of half a trillion dollars (similar to the entire economy of Sweden).

The Future of Jobs

The newest wave of machine learning and AI is unique in that it is affecting blue-collar as well as white-collar jobs. As a result, it is likely to have a significant effect on a very large number of jobs—47 percent of all jobs in the United States by some estimates. AI and machine learning are creating human-like cognitive capabilities in computer hardware and software. Applications of machine learning include natural language processing (speech recognition and translation), handwriting recognition, computer vision, facial recognition, and so on. Think of AI as the space inhabited by technologies such as machine learning, cognitive computing, and language processing. The combination and application of these technologies, such that they attempt to replicate or outperform our own cognition, could be seen as AI.

I can foresee a near-term future in which we will seamlessly interact with our computers using all five senses. This will make our computers more human-like, and perhaps more ubiquitous. After all, it is our senses that define us to the outside world. The keyboard will slowly disappear, as this is an artifact of a bygone era. What sets AI apart from prior technology waves is its ability to learn and then evolve and take on more complex tasks and decisions. It will evolve and grow *with us*.

As machine learning and AI evolve over the next few years, they will impact jobs and work in ways that were unimaginable a few years ago, replacing jobs that are predictable and repeatable. A combination of AI and robotics is likely to replace low-end, labor-intensive jobs such as picking fruits and vegetables, cooking food, or washing cars. A recent start-up company in San Francisco now has a robot serving as a barista. Many white-collar jobs will also be affected. We are already seeing the emergence of bots that carry out the work of financial and sports reporters and financial advisers. Machine learning systems are showing great promise in interpreting images in the fields of radiology and pathology.

Probably the most prolific product of the machine learning and AI revolution is the autonomous vehicle. The potential for the autonomous vehicle to cause disruption is massive. The transportation sector accounts for 7 percent of the U.S. economy. There are four million to five million individuals performing jobs in which driving is the major component. This includes truck drivers, bus drivers, and taxi drivers. A recent National Public Radio report pointed out that truck, delivery, and tractor driving is the most common occupation in 29 of the 50 U.S. states (Bui 2015). This does not even begin to touch on the secondary layer of jobs that are dependent on or associated with transport, including vehicle sales, insurance, mechanics and engineers, parking lots, retail, and fuel. This layer is one that is already feeling pressures from technological change. Furthermore, the annual economic cost of road traffic accidents in the United States alone is around \$242 billion (about 1.6 percent of our 2010 gross domestic product). All of this translates into jobs.

Are We Doomed to a Jobless Future?

I am not a pessimist. Historically, economies have always grown. Old jobs disappear and new jobs appear. Existing industries are transformed or die and new industries are created. Schumpeter's creative destruction in action! While many job classes are beginning to shrink, we are already witnessing the creation of many new job classes. Within the last 10 years, there has been an emergence and significant growth of new occupations including application developers, social media and digital marketing, User Experience (UX) designers, advanced manufacturing specialists, genetic counselors, and drone operators, to name a few. And the numbers are significant. There are an estimated 8 million to 10 million application developers worldwide. The number of drones registered in the United States now exceeds one million, and more than 60,000 people have received drone pilot licenses. The U.S. Bureau of Labor Statistics forecast a 29 percent annual growth rate for genetic counselors between 2014 and 2024. I believe that over the next two decades, the principal job-related challenge will be for individuals who are caught in the disruption-related turbulence and have not been able to reskill themselves for this new world order.

Challenges for Public Administrators

As the AI juggernaut gains momentum, I foresee five primary challenges for public administrators that need attention.

Electronic Government Services

Thanks to exponential improvements in natural language processing, machine learning–based voice recognition systems last year achieved 95 percent accuracy, a tipping point that puts it on par with humans. In some cases, there are claims of 99 percent accuracy in low-noise environments. This makes natural language processing ready for prime time, shifting from selective use to large-scale consumer adoption. Voice is the new browser! It is more efficient, too. An average typist can type 40 words a minute; humans speak at 110 to 150 words per minute. The use of keyboards moved from typewriters to computers as that was the only feasible way to provide input to a “not so smart” and expensive computer that needed exact and precise instructions. The keyboard has outlived its life. Voice will be the dominant interface with the machines of the future.

Starting in the mid-1990s, governments began moving their services to the internet on the desktop. As this was taking place, the smartphone arrived, and these services are now being moved to mobile platforms in the form of apps. Today, a majority of traffic on the internet is via mobile phones. However, governments are still behind the curve in this transition to mobile apps. And now comes the challenge of voice enablement of these services. Most governments have a very limited number of skilled professionals who are capable of dealing with these new voice platforms. The government is not prepared. To further compound the challenge, the voice platforms are proprietary, and services have to be designed separately for each platform.

As governments do not possess these high-demand skills, this provides an opportunity for the private sector to step in. Done right, this could result in productive partnerships. Done poorly, this could lead to outsourcing, with government having little control of these services.

Jobs, Economy, and Social Safety Nets

As machine learning and AI gain adoption, preparations must be undertaken to absorb the unemployment consequences that will be created by the disruptions. AI machines and robots will take over many jobs, both menial and low-end knowledge work.

To prepare for this scenario, many communities are starting to experiment with the concept of universal basic income, where an individual is guaranteed a regular, livable, and unconditional sum of money from government. There is no requirement to look for work, and this would be independent of any other income. This idea dates back to the eighteenth century, but it is gaining new recognition in light of this new AI landscape.

I consider the issue of AI and unemployment an income distribution problem. In a rational society, it makes perfect sense for AI machines and robots to do all the mundane and repetitive work. In fact, these machines could hypothetically provide us with limitless abundance. Therefore, if gross domestic product is growing

and the population remains more or less the same, it is simply a question of having the political will to distribute income in a more egalitarian fashion. Another solution to this problem would be to reduce the workweek. Prior to the Industrial Revolution, a 70- to 80-hour workweek was common, as was child labor. What we will need is a modern-day Henry Ford, as he supported the idea of a 40-hour workweek. It would be interesting to see whether the owners of capital would be willing to redistribute income or face the prospects of a revolution.

Revenue Shortfall

A recent Deloitte Insights report stated that “the U.S. public sector will likely have to figure out how to offset anticipated declines in the \$251 billion annually generated from fuel taxes, public-transportation fees, tolls, vehicle sales taxes, municipal parking, and registration and licensing fees. All these revenues are tied to today’s reality of individually owned and operated vehicles—for instance, the need for parking diminishes with the rise of autonomous-drive shared mobility” (Corwin et al. 2015). And this is just from the transportation sector alone.

If anything, an aging society will need to provide an increasing amount of services, as well as safety net resources. An idea to cover this shortfall, which was originally suggested by Bill Gates, is to tax robots. The European Union also considered legislation to tax robots in order to fund worker retraining but ultimately rejected it. Last October, the government of Saudi Arabia granted citizenship to a robot name Sophia, the first robot to receive citizenship in any country. One would imagine that if a robot can be a citizen, the privilege of paying taxes comes along with it. Incidentally, last November, Sophia was also named the United Nations Development Programme’s Innovation Champion, the first nonhuman to be given a UN title.

Consumer Protection

There is prevalent assumption that AI will be a neutral entity—an equalizer. This is proving not to be the case. AI machines reflect the bias of their creators or societal biases embedded in the data used to train the machines. AI learns from data. Since our data carry current biases, these machines will only institutionalize such tendencies. Imagine you are trying to train an AI machine and showing it pictures of roofers, an occupation that is 98 percent male. Applying probabilistic models, the system will be significantly biased in assuming that all roofers are male. The same could be said for speech-language pathologists, 98 percent of whom are female. Similar biases exist in most historical data along the lines of gender, ethnicity, and other demographics.

According to Tom Simonite, writing for *Wired* magazine and describing research by Professor Vicente Ordóñez,

Their results are illuminating. Two prominent research-image collections—including one supported by Microsoft and Facebook—display a predictable gender bias in their depiction of activities such as cooking and sports. Images of shopping and washing are linked to women, for example, while coaching and shooting are tied to men.

Machine-learning software trained on the data sets didn’t just mirror those biases, it amplified them. If a photo set generally associated women with cooking, software trained by studying those photos and their labels created an even stronger association.

Since it would be a rare case that the machine learning data set is neutral, the machines would likely end up amplifying the bias or perhaps creating new biases of their own that we cannot even predict. Let us not forget the ability of machine learning to see patterns that are indiscernible to us mere humans.

Another source of AI bias is algorithms. Algorithms are rules, some of which provided by humans and some of which are machine generated, that are embedded in AI machines. Some of these algorithms are becoming so complex that no one fully understands them, to such an extent that a proposal has been put forward to create an AI machine that is capable of understanding another machine’s algorithms!

A 2017 report from AI Now Institute recommended the following:

Core public agencies, such as those responsible for criminal justice, healthcare, welfare, and education (e.g., “high stakes” domains) should no longer use “black box” AI and algorithmic systems. This includes the unreviewed or unvalidated use of pretrained models, AI systems licensed from third party vendors, and algorithmic processes created in-house. The use of such systems by public agencies raises serious due process concerns, and at a minimum such systems should be available for public auditing, testing, and review, and subject to accountability standards.

A system currently being used by courts and corrections departments illustrates the risk of bias in AI systems. This system is used to assess a defendant’s risk of committing another crime or the likelihood of skipping a court date. There is considerable controversy as to whether these AI systems are biased against African Americans and other minorities as a result of biases in the training data.

Privacy

It is now the norm that any electronic service provider collects data resulting from its interactions or transactions with users and subsequently uses such data to extract value. There is no free lunch! It takes resources to provide services offered by companies such as Facebook, Google, and Twitter. Their business models essentially involve direct advertising as well as the collection and use of personal and transactional data for marketing purposes. These data, coupled with machine learning and AI, offer these companies an unprecedented capacity to profile individuals. The net result is a complete loss of personal privacy. For starters, one ought to have the right to opt out or, better yet, to opt in for the capture of personal data. This would have a significant impact in preserving privacy in the digital age. The European Union has taken a bold step with the General Data Protection Regulation (GDPR), which is designed to enable individuals to better control their personal data. Many assert that the GDPR may be an overreaction to this growing problem.

It is now well established that data are an asset because they have a market value. Unlike physical assets, data are different. They can have unlimited copies. The next decade will see questions not just about who owns these data and what rights we have as individuals over them, but also what our fundamental relationship with data will look like in a connected world.

Conclusion

The pace of change *outside* our public policy institutions is faster than the pace of change *within*. That is a problem. There is an urgent need for public administrators to get engaged *proactively*—they cannot simply wait for technology to shape the landscape before putting fences around it. Even the leaders of the tech industry—such as Elon Musk, Mark Zuckerberg, and many others—are calling for the government to be more engaged and more active in shaping policy around change.

We seem to ask questions of our technology that make us wonder, “Will AI be good or evil?” The real question ought to be, “Will *we* be good or evil?”

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